

8

CALCULUS I, QUIZ-4, 22/04/26 Name: TEJAS SAH

1. Write any three results you learned in this course (with all the hypotheses and conclusions).
2. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = 2x^3 - 3x^2 + 2y^3 + 3y^2$. Find the local extrema of f .

Q1:

Result 1: Heine-Borel Theorem:

$\hookrightarrow$ A set S in $\mathbb{R}^n$ is compact $\iff S$ is closed and bounded in $\mathbb{R}^n$.

Result 2:

A function f is continuously Differentiable $\iff$ All partial derivatives exist, and are continuous
(i.e, f' exists and is continuous)

~~Result 3: Inverse function theorem: $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $f \in C^1(X)$~~

Result 3: $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous on whole of $\mathbb{R}^n$

Inverse image of any open set in the image is open.

Result 4: Implicit function theorem:

$f: \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$, $f \in C^1(X)$

Given $\exists (a, b) \in X$ such that $f(a, b) = 0$ and $f'(a, b)$ has rank n $\rightarrow$ written as $\begin{bmatrix} A_x & A_y \\ n \times n & n \times m \end{bmatrix}$

Then, $\exists U \subseteq X$, $W \subseteq \mathbb{R}^m$ and a $C^1(X)$ function g such that
 $(a, b) \in U$, $b \in W$

$f(g(y), y) = 0, \forall y \in W$ and $g(b) = a$, and $g'(b) = -A_x^{-1} A_y$

Results: $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, if f is differentiable at a ,

then $f'(a) = \begin{bmatrix} D_i f_j(a) \end{bmatrix}_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}}$ matrix (where D_i corresponds to directional derivative along e_i)